

SI for Approximating spatial processes with too many knots degrades predictive accuracy of SPDE models

Table S1. Species common names, occurrence rates (of 702 tows), and catch per unit effort (kg per km²) in the 2018 WCBTS survey.

Common	Scientific	CPUE	Occurrence
Dover sole	<i>Microstomus pacificus</i>	3261.9	0.84
Sablefish	<i>Anoplopoma fimbria</i>	2007.0	0.62
Petrale sole	<i>Eopsetta jordani</i>	1042.1	0.41
Lingcod	<i>Ophiodon elongatus</i>	1166.7	0.32
Darkblotched rockfish	<i>Sebastes crameri</i>	1386.5	0.15
Pacific ocean perch	<i>Sebastes alutus</i>	8468.0	0.06

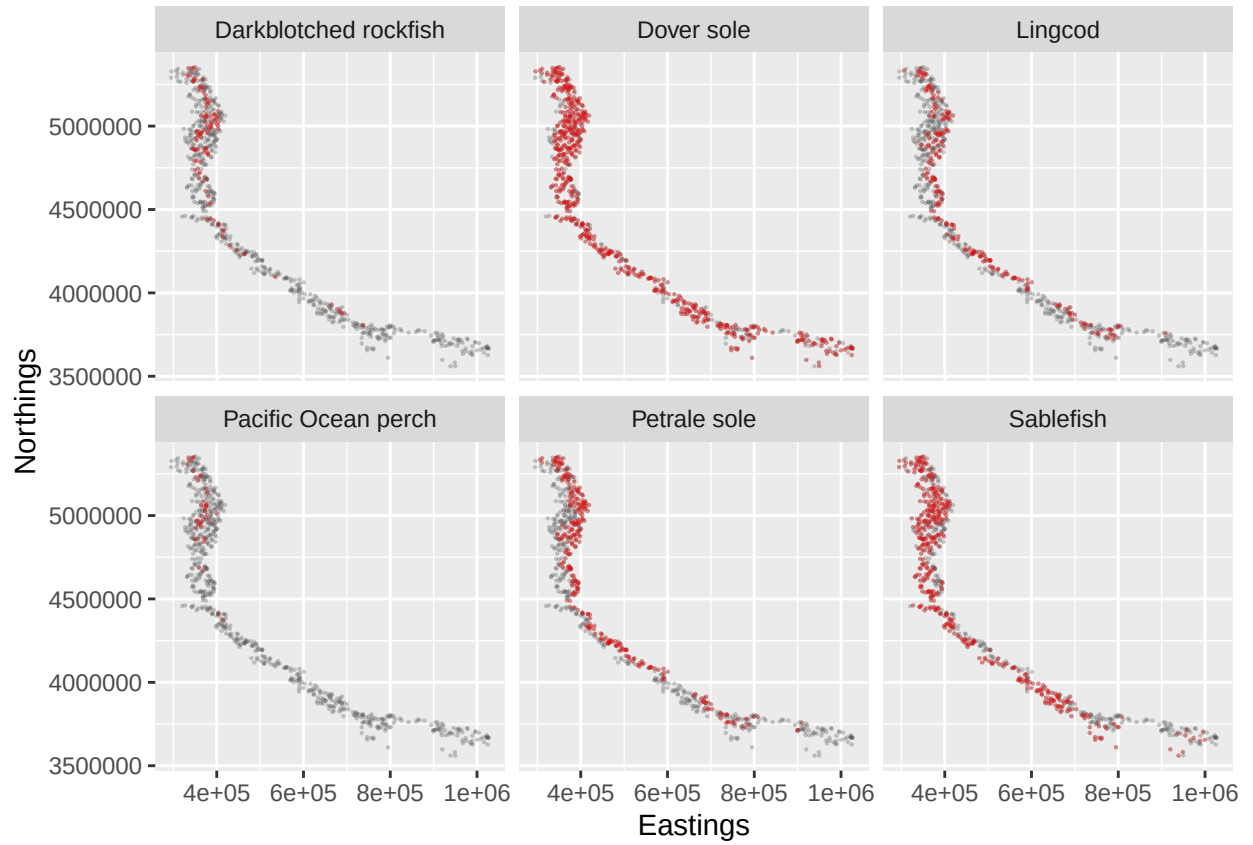


Figure 1: Distribution of presence-absence data in the 2018 WCBTS dataset. Grey points represent absences, red points represent occurrences

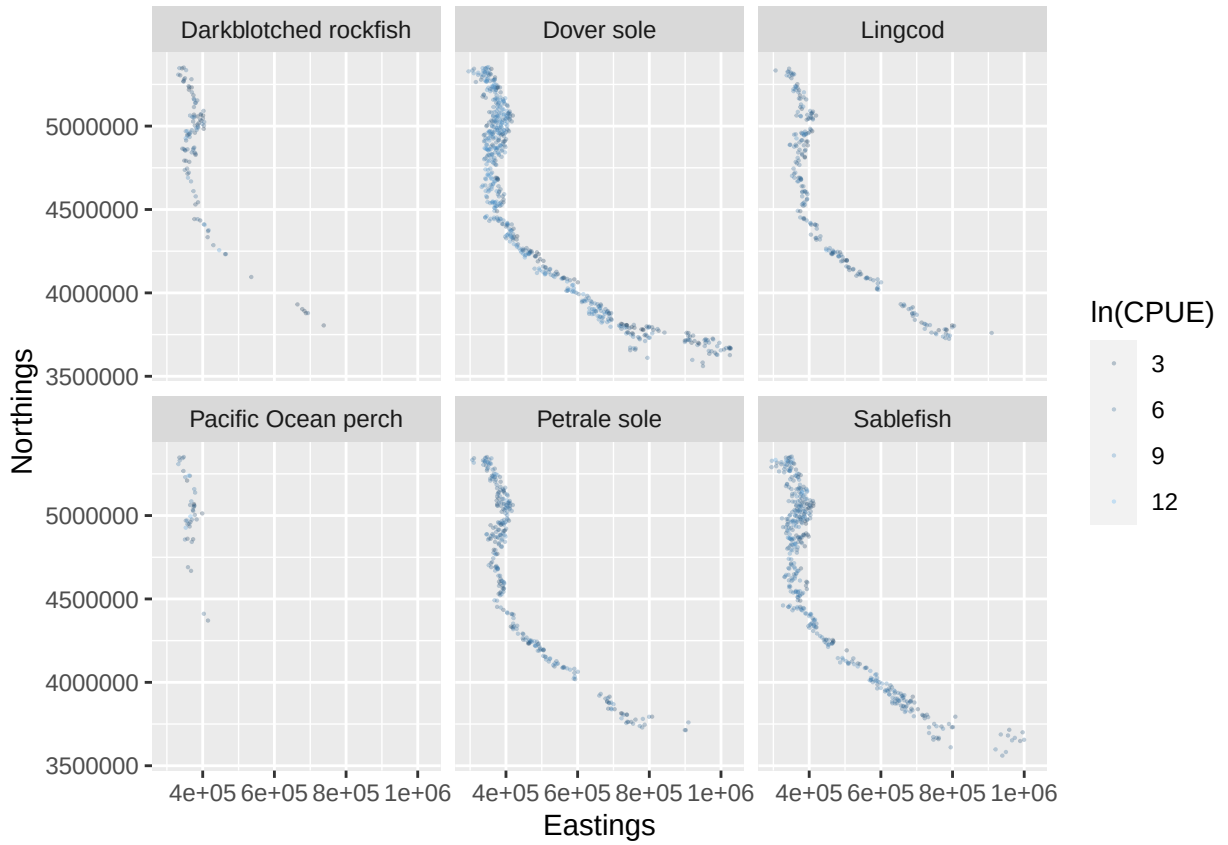


Figure 2: Catch per unit effort (kg per km²) in the 2018 WCBTS survey. Estimates are shown in log space.

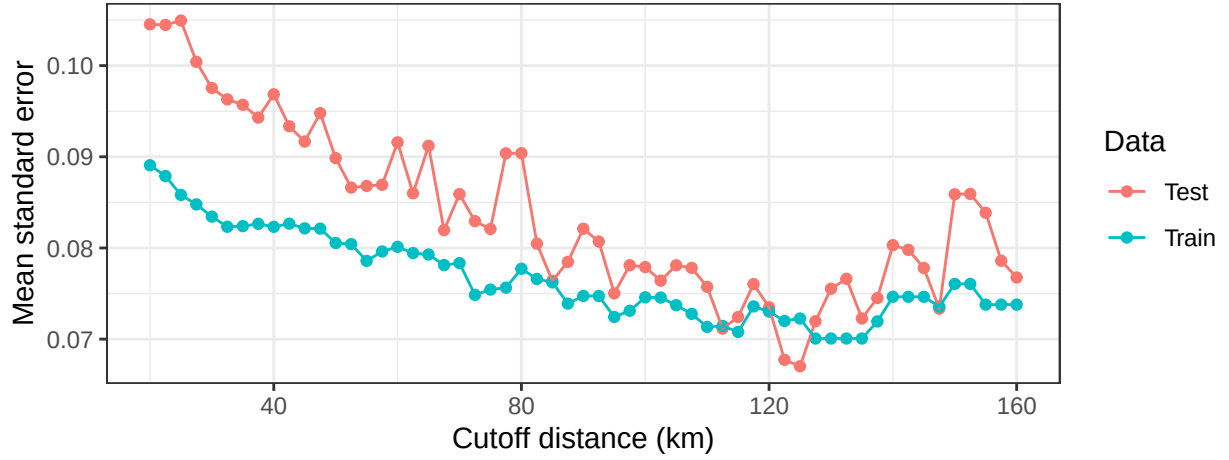


Figure 3: Mean standard errors of predictions for models of ocean temperature in the WCBTS survey, June 2018. Mean standard errors are calculated across all data points in the respective set ($n = 150$ train, 25 test).

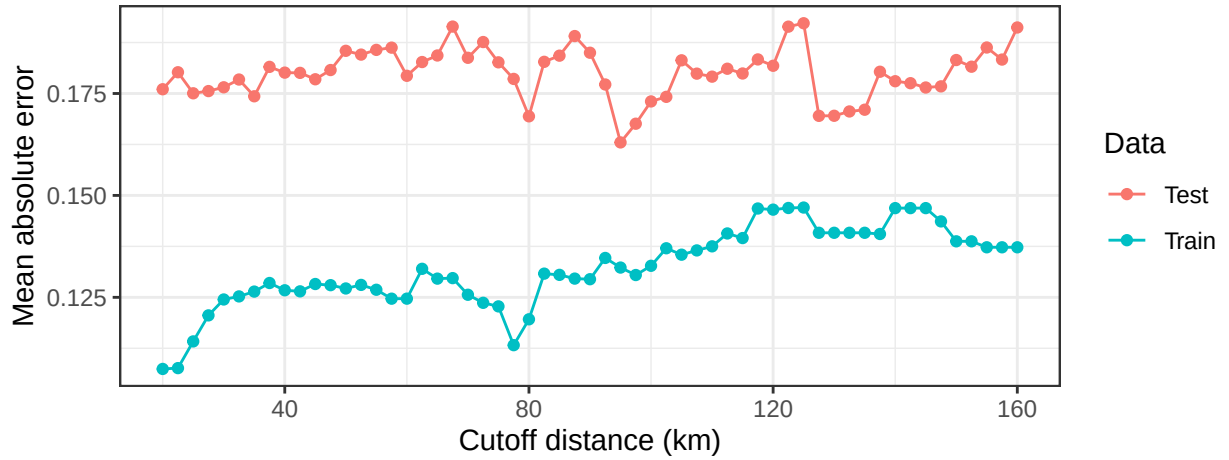


Figure 4: Mean absolute errors of predictions for models of ocean temperature in the WCBTS survey, June 2018. Mean absolute errors are calculated across all data points in the respective set ($n = 150$ train, 25 test).

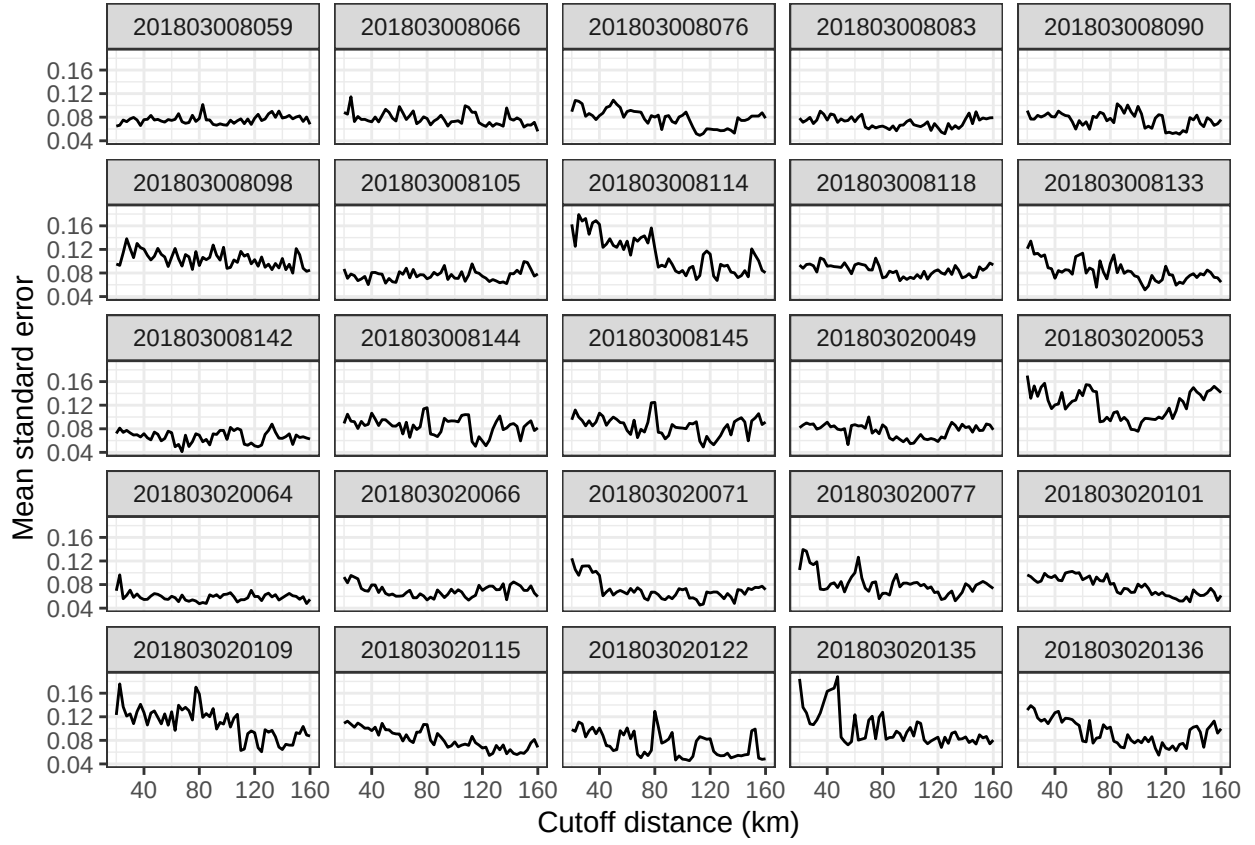


Figure 5: Mean standard errors of predictions for models of ocean temperature in the WCBTS survey, June 2018. Estimates are shown for each held out trawl in the test set.

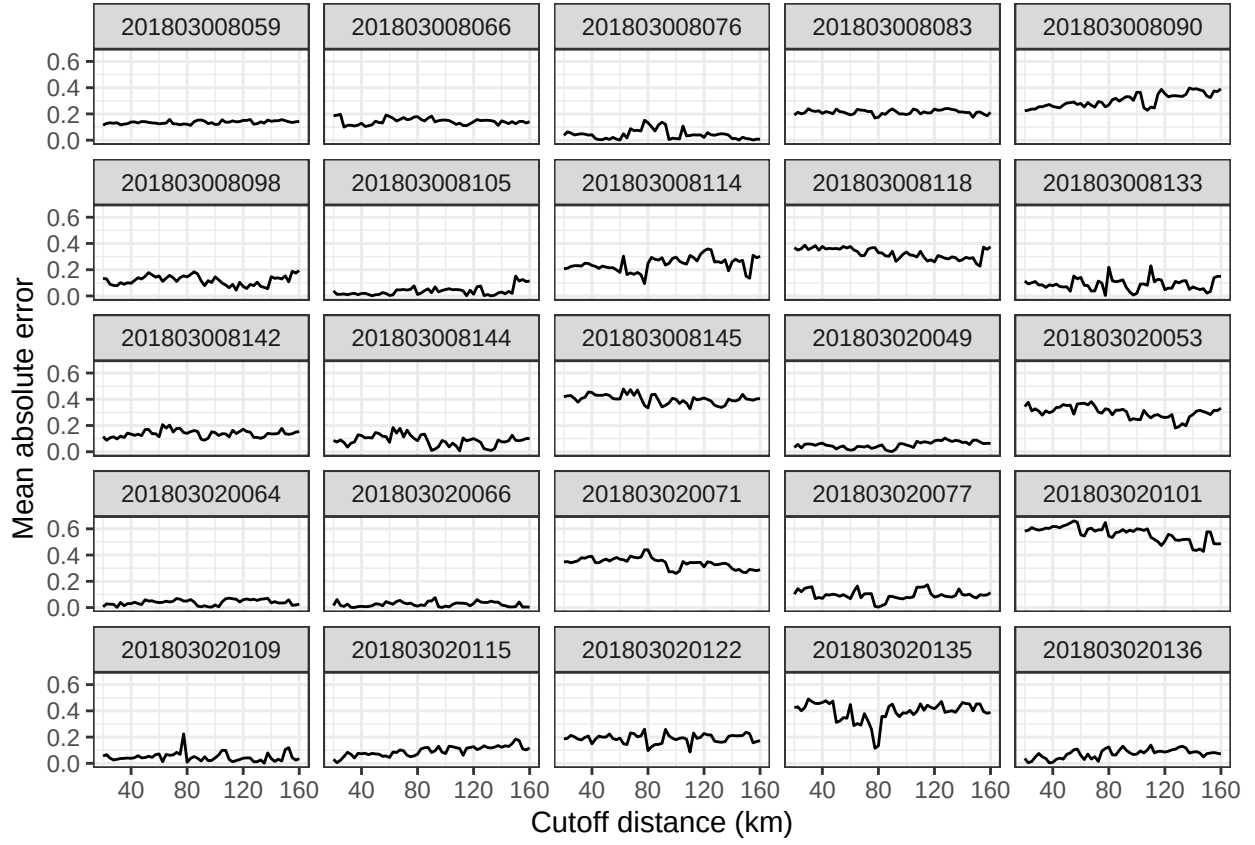


Figure 6: Mean absolute errors of predictions for models of ocean temperature in the WCBTS survey, June 2018. Estimates are shown for each held out trawl in the test set.

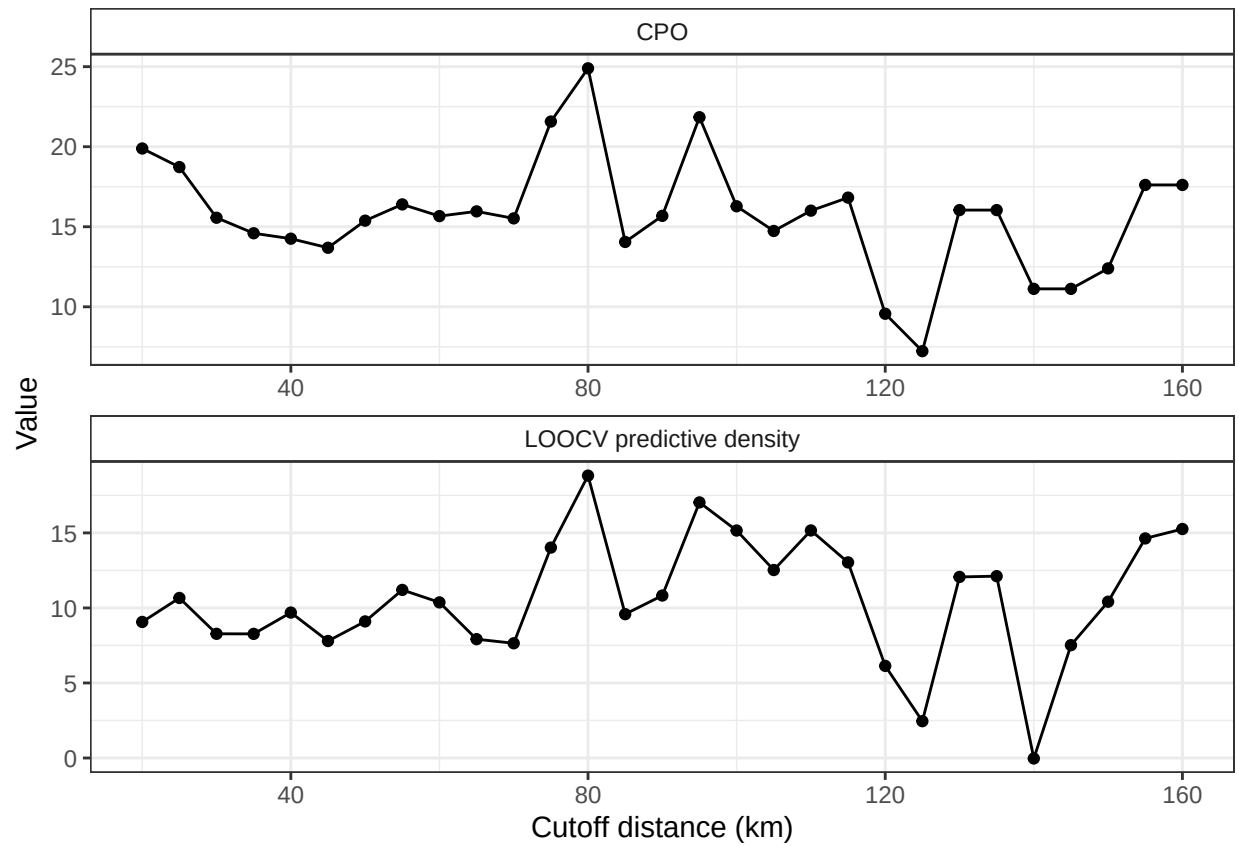


Figure 7: Predictive density calculated with Leave One Out Cross Validation (LOOCV) and the conditional predictive ordinate (CPO), using sampled temperature from the June 2018 WCBTS survey. Estimates are shown for a subset of cutoff distances evaluated (x-axis).

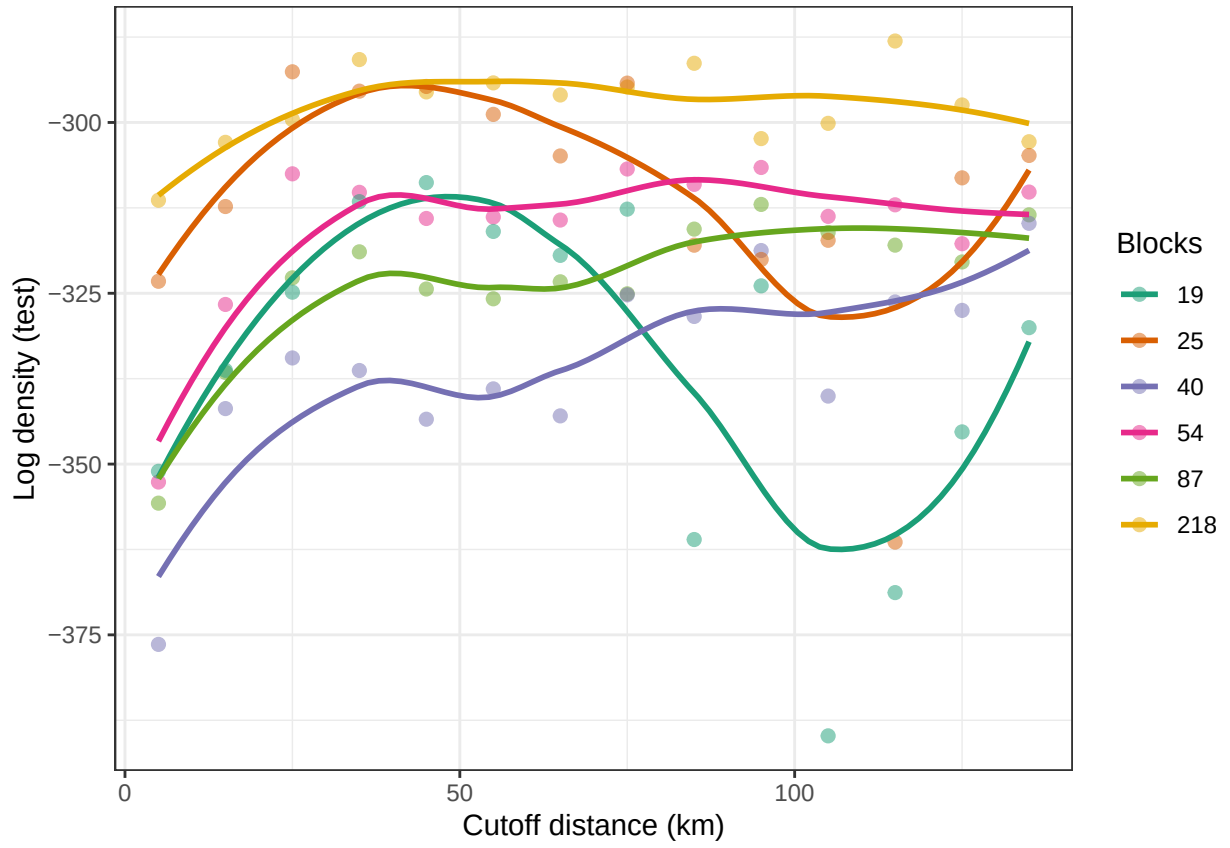


Figure 8: Out of sample predictive density for Dover Sole presence-absence models, as a function of cutoff distance (km) and number of spatial blocks used for 10-fold spatial cross-validation. Cutoff values between 5-135km are used across all scenarios.

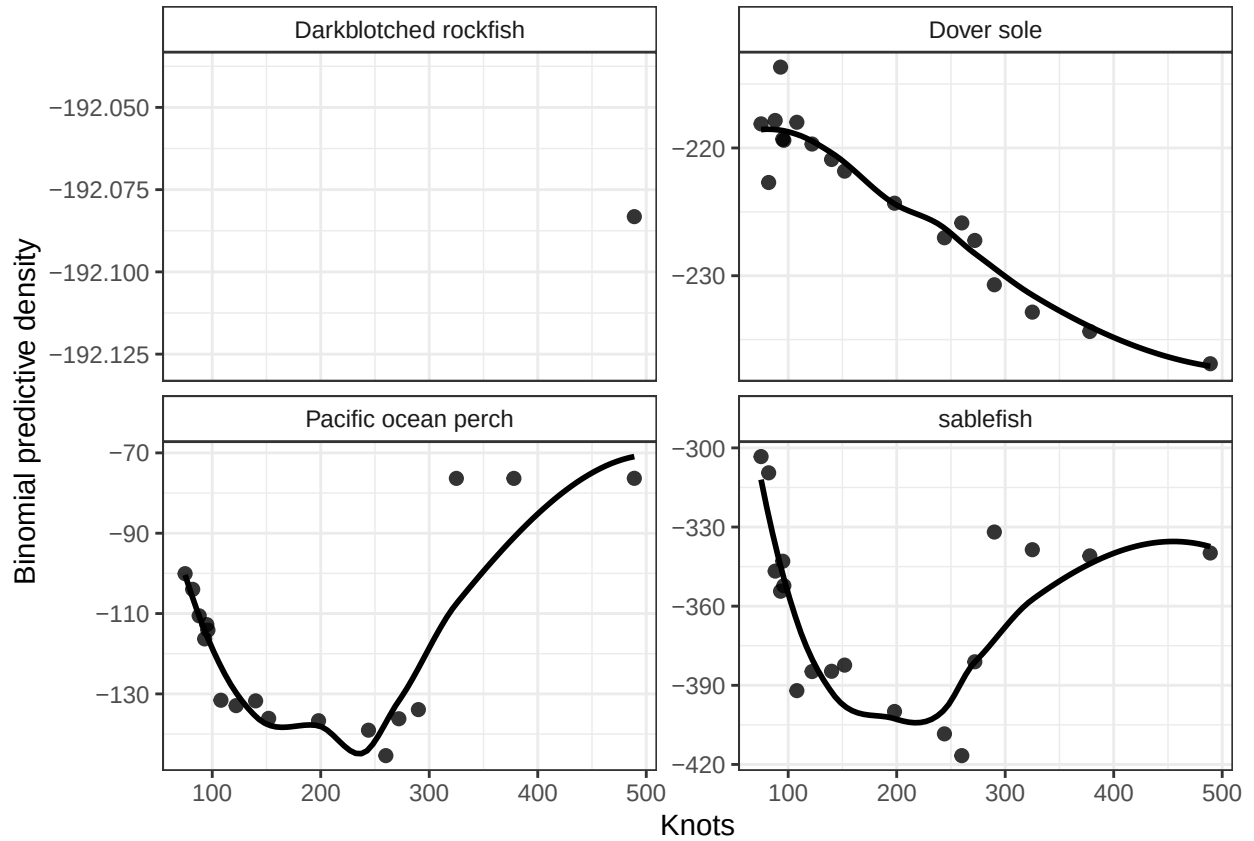


Figure 9: Binomial predictive densities for four species from the WCBTS survey, as a function of number of knots (using cutoff values from 3-120km). Estimation is done using Template Model Builder (TMB) instead of INLA.

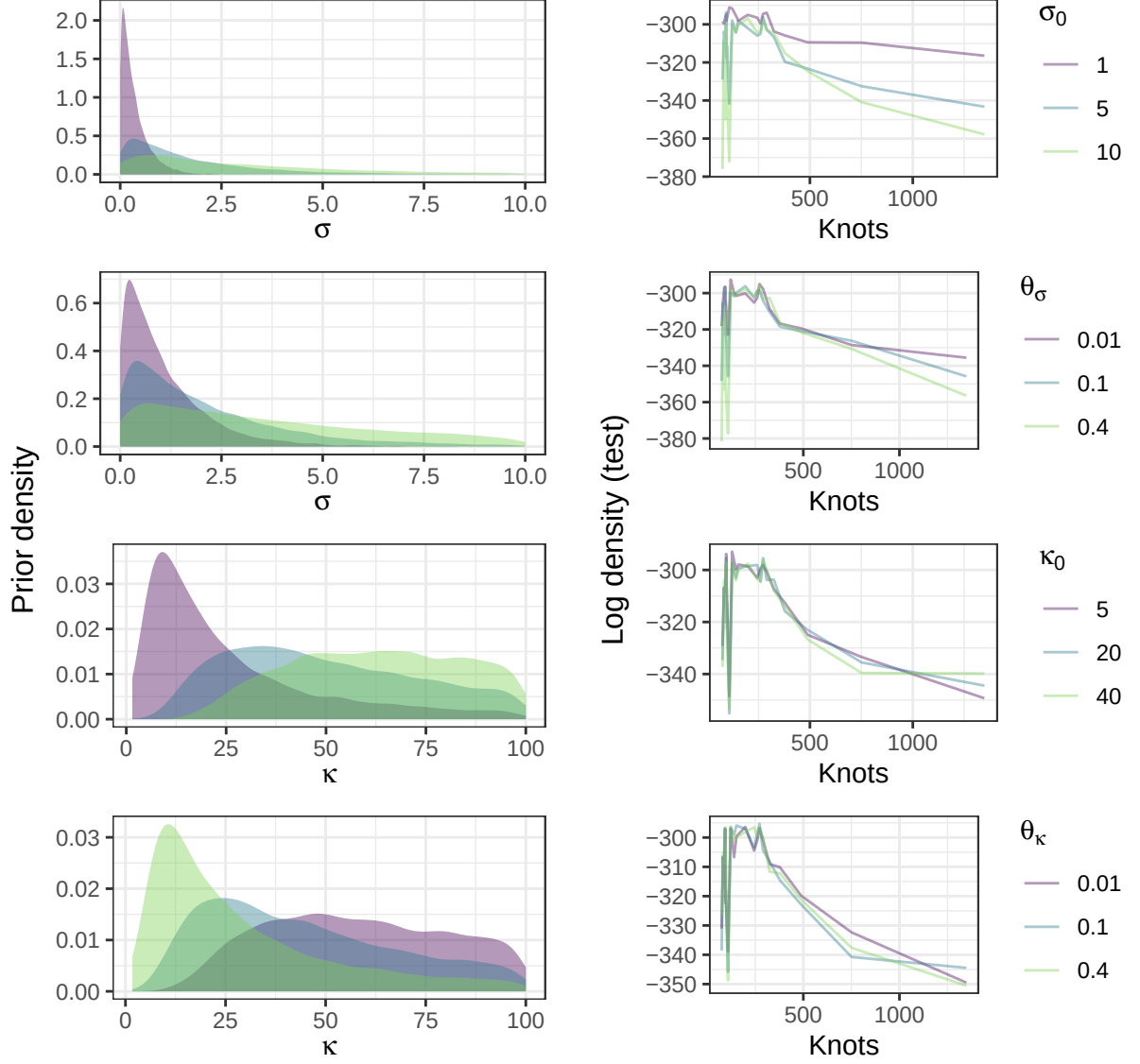


Figure 10: Prior sensitivity for Dover sole presence-absence model. The left column represents the marginal prior of the Matérn standard deviation or range, as a function of 3 hyperparameters. The right column presents the out of sample log predictive density (binomial), using cutoff values 3–120km.